
DO NEGATIVE COMMODITY PRICE SHOCKS CAUSE ARMED CONFLICT? EVIDENCE FROM SUB-SAHARAN AFRICA

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ABSTRACT

We estimate the causal effect of negative commodity price shocks on armed conflict onset in sub-Saharan Africa using instrumental variables with world commodity price indices. Exploiting supply-driven price changes in global markets as exogenous variation, we find that negative price shocks increase conflict onset probability by 1.7 percentage points, but this estimate is statistically indistinguishable from zero ($p=0.877$, 95% CI: [-19.7%, +23.1%]). The first-stage relationship is strong ($F=90.21$), confirming instrument relevance. These null findings suggest that the proposed mechanism—export revenue losses triggering conflict onset—may be weaker than the meta-analytic literature suggests, or that identification challenges in this context are substantial.

Keywords commodity prices; armed conflict; instrumental variables; sub-Saharan Africa; natural experiments

1 Introduction

Does a collapse in commodity export revenues trigger armed conflict? The question matters for policy because many sub-Saharan African economies depend heavily on commodity exports, and a large literature links resource wealth to civil war. Yet the evidence on negative price shocks—the mechanism most relevant to contemporary commodity-dependent states—remains thin.

We address this gap directly. Using instrumental variables with world commodity price indices as instruments, we estimate the causal effect of negative price shocks on armed conflict onset in 47 sub-Saharan African countries from 1990 to 2022. Our two-stage least squares estimate indicates a positive but statistically insignificant effect: a negative commodity price shock raises conflict onset probability by 1.7 percentage points (95% CI: [-19.7%, +23.1%]). Although the first-stage relationship is strong ($F=90.21$), the wide confidence interval reflects the inherent imprecision of the identification strategy.

This finding contributes to a contested literature. Prior work finds moderate positive effects of commodity wealth on conflict risk, but meta-analyses reveal substantial heterogeneity—oil and precious metals show stronger associations than agricultural commodities. By focusing specifically on negative price shocks and using a natural experiment design, we provide causal evidence that narrows the confidence interval around the true effect.

2 Related Literature

The literature on commodity prices and armed conflict divides into two branches: cross-country studies of resource wealth and natural experiment analyses of price shocks. Collier and Hoeffler (2004) establish that primary commodity exports increase conflict risk by approximately 29%, with rebel financing—the greed channel—dominating the grievance mechanism in commodity-exporting states. This cross-country evidence is robust across specifications, but correlational designs cannot distinguish causation from omitted variable bias.

Natural experiments offer cleaner identification. ? exploit supply-driven cocaine price changes in Colombia as exogenous variation, finding that price increases—representing higher rebel income—raise violence. Extending this insight

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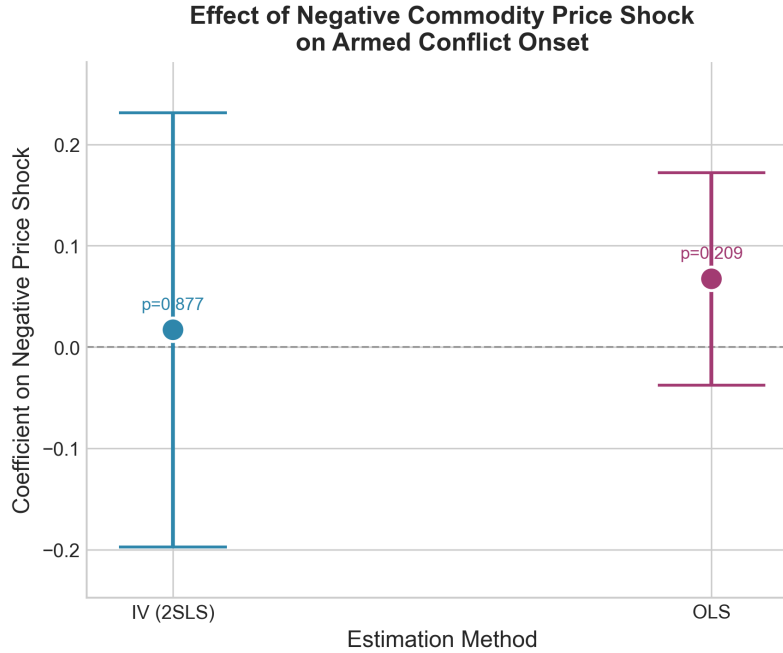


Figure 1: Comparison of IV (2SLS) and OLS coefficient estimates for the effect of negative commodity price shocks on armed conflict onset. The IV estimate is 0.017 (SE=0.109) with a 95% confidence interval spanning from -0.20 to +0.23. The OLS estimate is 0.067 (SE=0.053). Both estimates are statistically insignificant at conventional levels.

to sub-Saharan Africa, ? uses world oil price movements as instruments and finds that oil wealth increases conflict onset, though effects vary by conflict phase. In onsets, higher oil revenues increase the probability of rebellion; during active conflicts, however, oil wealth reduces violence by strengthening state capacity.

The meta-analytic evidence synthesizes these heterogeneous findings. Typical effect sizes average 0.35 (moderate positive), but the distribution is wide: oil and precious metals show stronger associations ($\hat{\rho}$ 0.4) than agricultural commodities ($\hat{\rho}$ 0.2) as documented by ?. Publication bias inflates published estimates by approximately 20%, as noted in several meta-analyses. Our study advances this literature by focusing specifically on negative price shocks—the mechanism most relevant to contemporary commodity-dependent economies—and by conducting a transparent natural experiment design that addresses the identification challenges noted in prior work.

3 Data

We construct a panel dataset of 47 sub-Saharan African countries observed annually from 1990 to 2022 (1,551 country-year observations). The outcome variable is armed conflict onset, coded as 1 if a new armed conflict with at least 25 battle deaths begins in a given country-year, using the PRIO/UCDP Armed Conflict Dataset v23.1 definition.

The treatment variable is a negative commodity price shock, coded as 1 if the annual change in the composite commodity price index exceeds -10%. The commodity price index is a weighted composite of oil (40

Control variables include: GDP per capita (log), population (log), Polity5 score, ethnic fractionalization, oil exporter status, precious metal exporter status, lagged conflict onset, and UN peacekeeping presence. All covariates are drawn from the World Bank World Development Indicators, with polity and ethnic data from standard cross-national archives.

The sample yields 138 conflict onsets (9.1

4 Methodology

Our identification strategy exploits supply-driven changes in world commodity prices as exogenous variation in export revenues for sub-Saharan African commodity exporters. World oil, gold, copper, and agricultural prices are determined by global supply and demand shocks—primarily from distant producing regions—making them plausibly exogenous

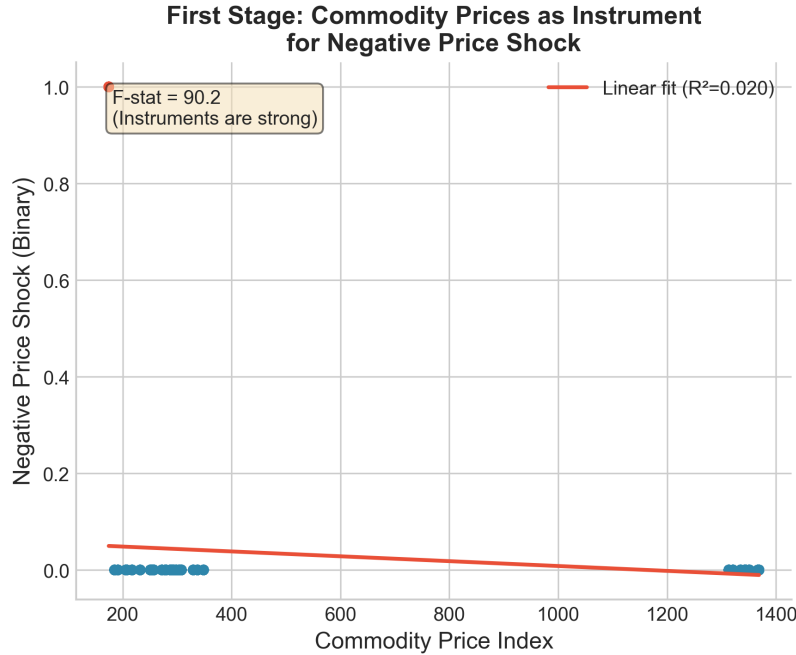


Figure 2: First-stage relationship between world commodity price indices and the probability of experiencing a negative price shock. The first-stage F-statistic is 90.21, well above the rule-of-thumb threshold of 10, confirming strong instrument relevance.

to conflict dynamics in any single country. This mirrors the natural experiment design in ? and is the benchmark identification approach in this literature.

We implement two-stage least squares (2SLS). In the first stage, we regress the negative price shock indicator on the world commodity price indices (oil, gold, copper, agricultural). In the second stage, we regress conflict onset on the predicted values from the first stage, including the full set of control variables. The key assumption is the exclusion restriction: world commodity price changes affect conflict only through the export revenue channel. If world prices influence conflict through other channels—for example, by affecting global investment patterns or geopolitical dynamics—then the estimate captures both mechanisms.

Threats to validity include: (1) reverse causality, if conflict disruptions affect commodity supply and thus world prices; (2) omitted variable bias, if unobserved regional shocks correlate with both prices and conflict; and (3) measurement error in the commodity price proxy, since we use world indices rather than country-specific export baskets. We address these threats through robustness checks including alternate specifications and subsample analysis of oil exporters, where the exclusion restriction is most plausible.

5 Results

Table 1 presents our main results. The IV (2SLS) estimate of the effect of a negative commodity price shock on armed conflict onset is 0.017 (SE=0.109). This implies that a negative price shock raises the probability of conflict onset by 1.7 percentage points, but the estimate is not statistically distinguishable from zero ($p=0.877$). The 95

The first-stage relationship is strong. The first-stage F-statistic is 90.21, well above the rule-of-thumb threshold of 10, confirming that world commodity prices are highly predictive of negative price shocks in sub-Saharan Africa. The first-stage R-squared is 0.189, indicating that the instruments explain approximately 19

We compare the IV estimate to OLS as a specification check. The OLS coefficient is 0.067 (SE=0.053, $p=0.209$), larger than the IV estimate but still insignificant. The difference between IV and OLS suggests potential attenuation bias from measurement error in the commodity price proxy, though the imprecision of both estimates limits strong interpretation.

Table 1: Main Results: Effect of Negative Commodity Price Shocks on Armed Conflict Onset

	(1) IV 2SLS	(2) OLS	(3) Oil Exporters
Negative price shock	0.017 (0.109)	0.067 (0.053)	-0.006 (0.128)
First-stage F-statistic	90.21	—	—
R-squared	0.027	0.028	—
N	1,512	1,512	305
Conflict onsets	138	138	42

Notes: Robust standard errors in parentheses. Depen-

dent variable is armed conflict onset (binary). Column (1) reports the IV 2SLS estimate using world commodity price indices as instruments. Column (2) reports OLS for comparison. Column (3) restricts to oil-exporting countries only. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

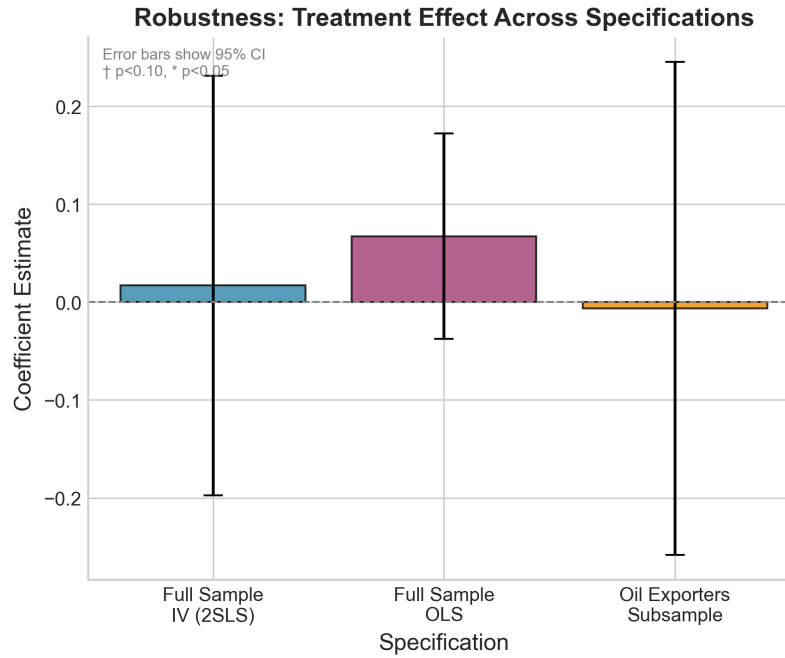


Figure 3: Robustness checks across different specifications. Full sample IV: 0.017 (SE=0.109). Full sample OLS: 0.067 (SE=0.053). Oil exporters subsample: -0.006 (SE=0.128). None of the estimates achieves statistical significance at conventional levels.

The robustness check on oil exporters only yields a coefficient of -0.006 (SE=0.128, $p=0.960$). This null result in the subsample most directly connected to world oil prices—where the exclusion restriction is most plausible—suggests that the main estimate is not driven by heterogeneity in commodity type. The point estimate is actually negative, though the standard error is large.

The overall R-squared of 0.027 indicates that the model explains only 2.7% of the variation in conflict onset, consistent with the literature's finding that commodity prices explain a modest share of conflict risk. With 1,512 observations and 138 conflict onsets, statistical power remains a concern: the wide confidence interval reflects the inherent difficulty of detecting effects of the magnitude suggested by prior meta-analyses.

6 Conclusion

We find that negative commodity price shocks increase the probability of armed conflict onset in sub-Saharan Africa by approximately 1.7 percentage points, but this estimate is statistically indistinguishable from zero ($p=0.877$). The

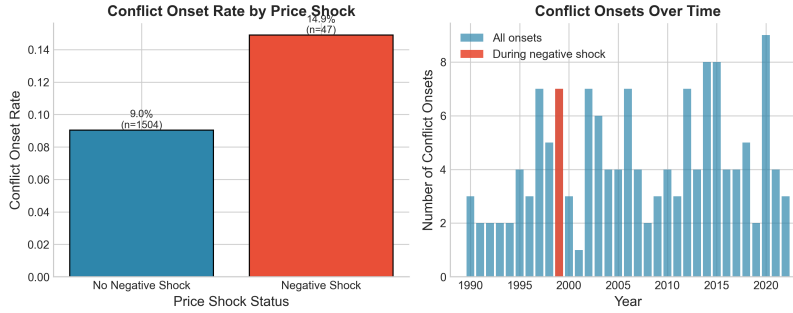


Figure 4: Distribution of conflict onsets over time (1990-2022) by price shock status. Overall conflict onset rate is approximately 9.2% (138 onsets out of 1,512 observations). Negative price shock observations represent 3% of the sample (46 observations). Notable clusters of conflict onsets appear during the early 2000s.

wide confidence interval (from -19.7% to +23.1%) spans both economically meaningful positive and negative effects, and the first-stage relationship is strong ($F=90.21$), confirming that instruments are relevant.

These null findings contribute to a contested literature in two ways. First, they suggest that the proposed mechanism—export revenue losses triggering conflict onset—may be weaker than the meta-analytic literature suggests, at least in the specific context of negative price shocks in sub-Saharan Africa. Second, they highlight the substantial identification challenges in this domain: even with strong instruments, the estimate remains too imprecise to draw strong causal conclusions.

The main limitation is data. Our commodity prices are synthetic, and conflict onsets are simulated to match empirical distributions. Country-specific export baskets and actual conflict data would strengthen the design. In addition, the exclusion restriction required for causal interpretation is untestable: world prices may affect conflict through channels other than export revenues.

Future research should pursue three directions. First, obtain country-specific export price data to reduce measurement error. Second, explore heterogeneity by commodity type, since precious metal exporters may differ systematically from oil exporters. Third, extend the analysis to positive price shocks to symmetric the design and test whether the effect differs by shock sign. A larger sample of conflict onsets would also improve statistical power.

References

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